WINXON NGUYEN
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EDUCATION

University of California-Los Angeles, Los Angeles, CA June, 2028
• *BS in Applied Mathematics w/ Specialization in Computing | BS in Statistics & Data Science*
• **Relevant Coursework***: Abstract Linear Algebra, Honours Physics, Discrete Structures, Differential Equations, Real Analysis, Data Structures & Algorithms, Numerical Analysis, Machine Learning*

EXPERIENCE & PROJECTS

CruX Neuroscience, *Developer*, Los Angeles, CA 2025-Present
• Built an end-to-end **ETL pipeline** in **Python** processing live 16-channel EEG data into a real-time focus-state classifier; framing the task as binary classification enabled threshold-based decision rules that reached 99% accuracy across experimental sessions.
• Applied signal-processing techniques to cut EEG artifact/noise by ~79%, improving classifier reliability across multiple users.
• Exposed the classifier through a **Python API** integrated into users' existing workflows, cutting time-on-task by 60-80% versus baseline completion times reported on comparable public datasets.

UCLA: Math + Code + Art, Los Angeles, CA 2026-Present
Researcher
• Developed **Python** applications creating higher dimensional Voronoi tessellations in art and music. Built out visual **deep learning** model to measure art and music styles, and transfer art semantic across pieces.
• Deconstructed **Meta's Segment Anything Model (SAM) image encoder** with a faculty advisor to study its object-segmentation reasoning, then derived a custom Voronoi density-field equation that converts segmented images into multi-dimensional generative mosaics
• Researched pipeline for Image Segmentation, **Fourier Feature Mapping** and **Vision Transformer** process to optimize promptable segmentation.
Lead Developer
• Lead development of interactive app in **Python** for Oceanside Museum of Art. Allows users to turn any custom image into a Voronoi tessellation with variable number of centroidal nodes and image weights. Incorporated **Lloyd relaxation algorithm** to increase user control over their custom art piece.
• Integrated **Meta SAM image encoder** into the pipeline to allow users to prompt generalized image segments.
• Collaborated with team to create an algorithm for using photo collages to replace image pixels, allowing users to create mosaics of their memories, from public database images.

UCLA Startup Labs: Data Infrastructure for ML in Predictive Systems, *Founder,* Los Angeles, CA 2026
• Built a scraping and embedding pipeline combining a custom CNN (image) and an adapted **NLP model** (text) to map multi-modal social signals into a 4-dimensional semantic feature space.
• Backtested a downstream trading strategy on ~50 event-contract markets on Kalshi over a 4-month window, achieving a **Sharpe ratio of 1.38.**

C-VUSD AI Task Force, *Team Lead*, Walnut, CA 2024-2025
• Designed district-wide AI adoption framework for scalable use in education through policy and teaching psychology.
• Trained specialized **RNN agents** for personalized teaching and administered use. Distributed to 20,000+ users.

International Bacc., *Research Team Lead*, Walnut, CA 2024-2025
• Wrote a 30+ page paper on **Fourier Analysis** & **Wave Dynamic** applications in order to improve musical harmonics. Concluded imperial study of the harmonic composition of an optimal music tone across different classifications of instruments.
• Developed a companion app in **Python** and live pipeline to input audio into **custom FFT** signal processing pipeline.

Music Transcriber, *Developer*, Walnut, CA 2026
• Developed a music transcribing app that transcribes sounds and music audio into readable sheet music in **Python** and **JS**. End-to-end **ETL** pipeline implementing **FTT** for cleaning artifacts, **ASR** for lyrics, as well as training neural net to detect for genres and style indicators.

TECHNICAL SKILLS

• *Computer Languages:* C++, C, Python, SQL, R
• *Tools:* Pandas, NumPy, PyTorch, Data Pipeline, Regression, Time Series Analysis, ETL, Vision Transformers, CNNs
• *Languages:* Fluent: English, Vietnamese, Mandarin
• *Awards:* Math Department (’22, ’23, ’25), C-VUSD Robotics Award ’17, CIF Athletic Scholar Award ’25, International Scholar ’25, SoCal Field Band (Gold 2021, Bronze ’22, Silver‘24), Louis Armstrong Award ’25